

Week 4 Capstone Project — Sales Analytics

EDA, Regression, Dashboard Insights and Recommendations

This report completes the Week 4 capstone requirements: data cleaning and EDA, a regression model for sales prediction, dashboard-ready business metrics, and stakeholder recommendations.

Metric	Result
Total sales	\$2,297,200.86
Total profit	\$286,397.02
Orders	5,009
Customers	793
Profit margin	12.47%
Returned orders	296 (5.91%)
Model R ²	0.260
Model MAE	\$189.91
Model RMSE	\$503.26

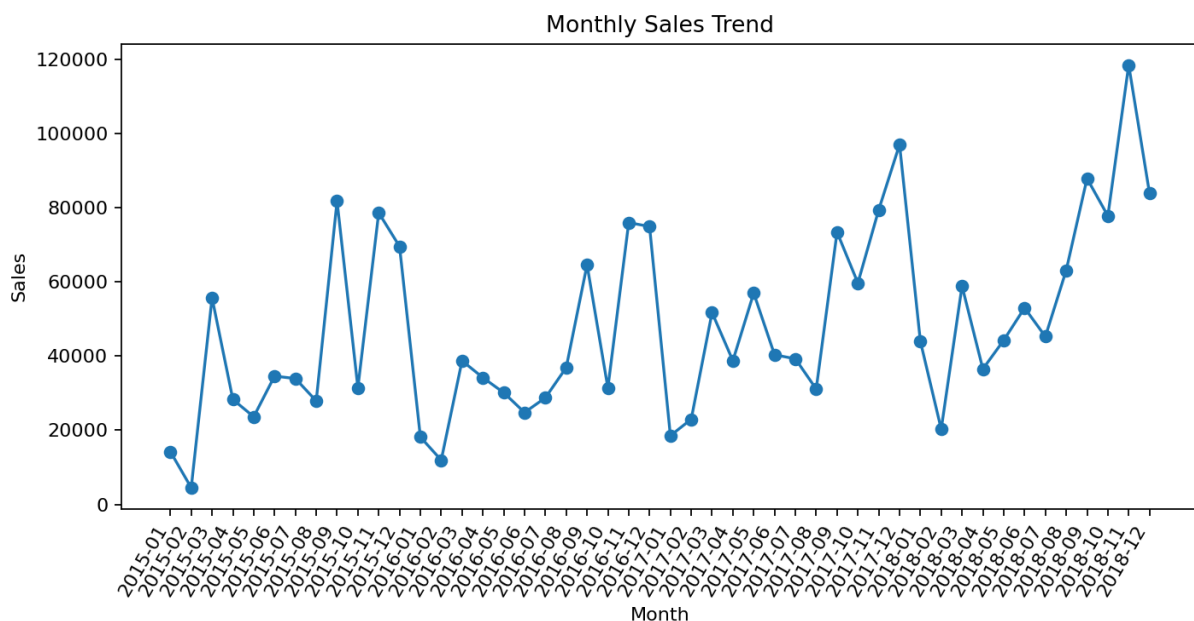
1. Data Cleaning

The working CSV contains 9,994 transaction records followed by a return-order lookup. The analysis separates those sections, converts dates and numeric fields, creates calendar features, flags returned orders, removes duplicates and excludes rows without a usable sales target or order date.

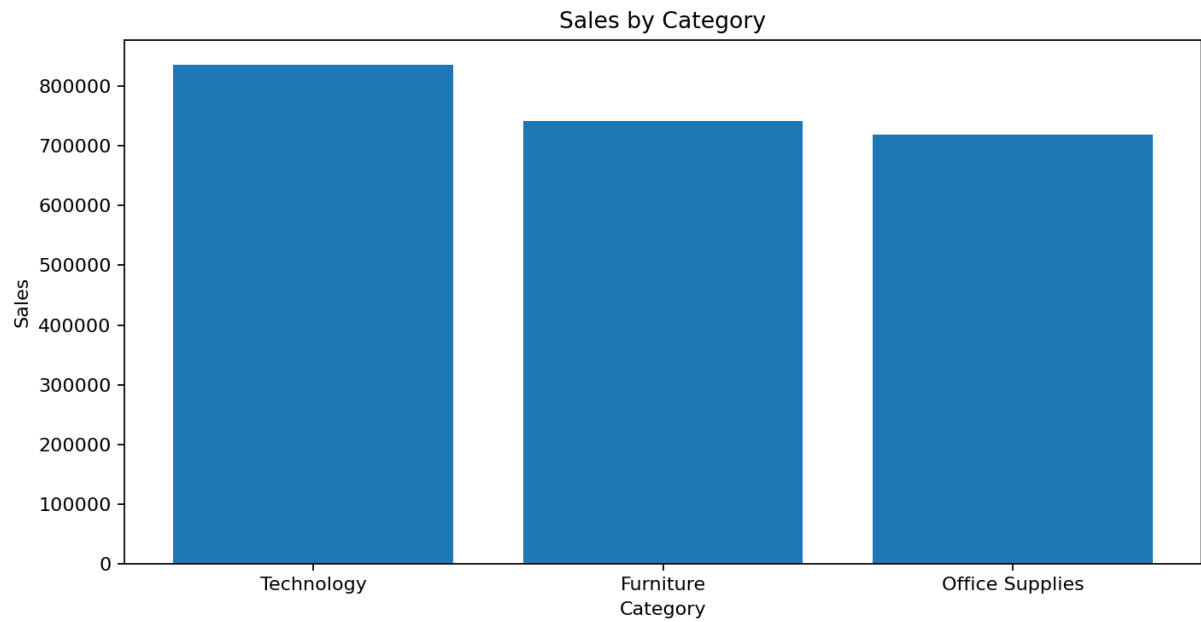
2. Exploratory Findings

Technology generated the highest sales at \$836,154. The West region produced the highest profit at \$108,418. Tables were the weakest sub-category by profit at \$-17,725. November 2018 was the strongest month at \$118,448 in sales.

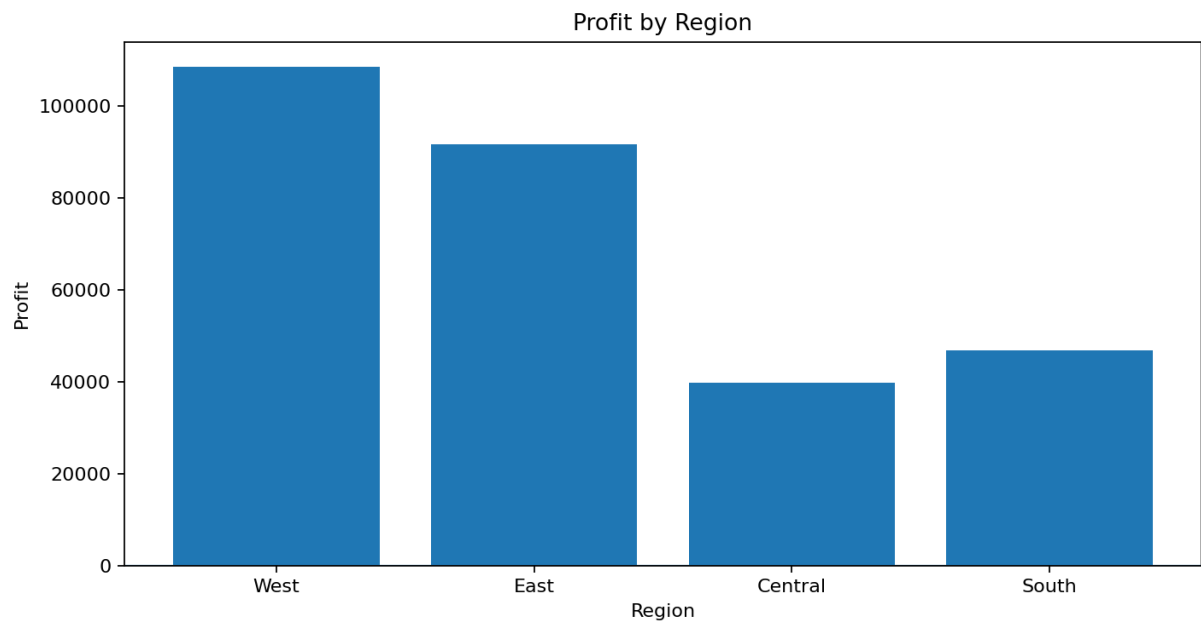
Monthly sales trend



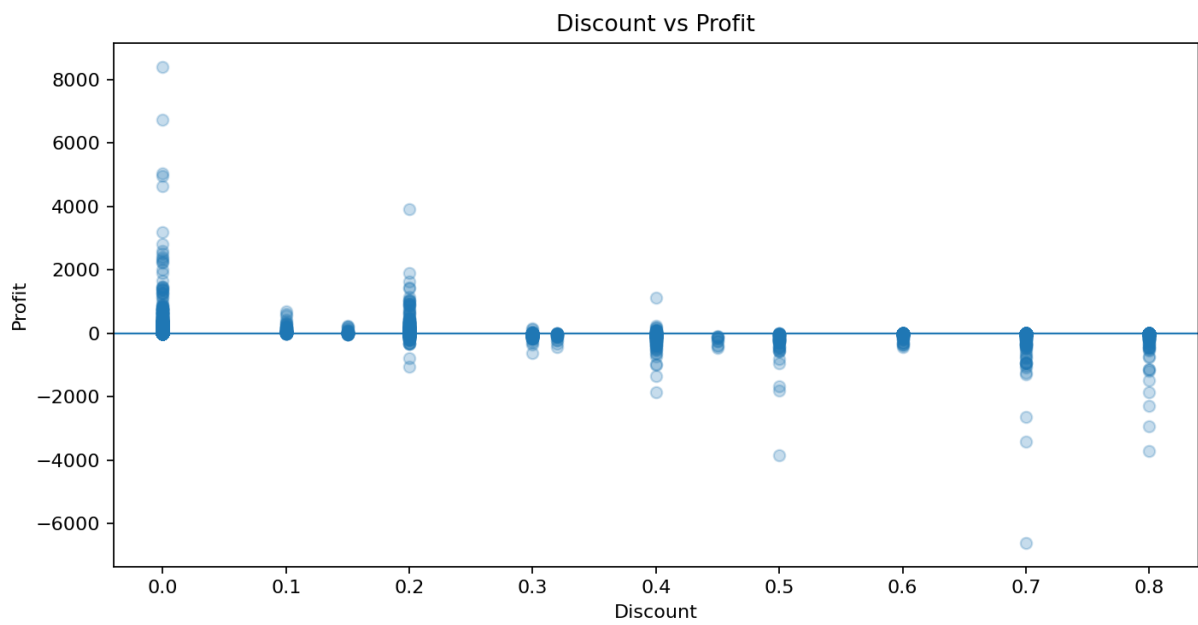
Sales by category



Profit by region

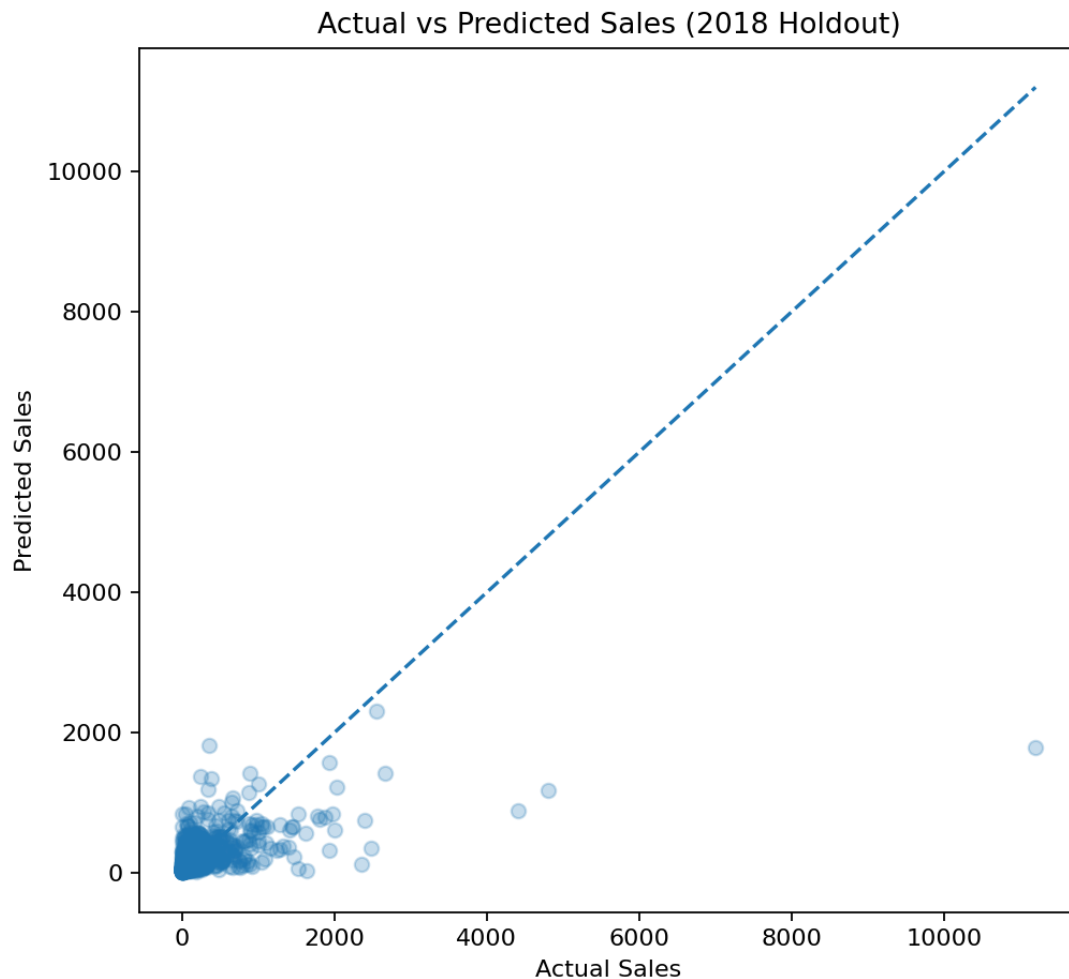


Discount vs profit



3. Sales Regression Model

A Random Forest regression model predicts sales using quantity, discount, calendar features, shipping mode, segment, region, category and sub-category. Profit is excluded to avoid target leakage. The model is trained on 2015–2017 and evaluated on 2018.



Evaluation: MAE \$190, RMSE \$503, R^2 0.260. This is a useful baseline, but additional marketing, customer-history and inventory variables would likely improve prediction quality.

4. Recommendations

Protect Technology category momentum; review pricing and discounting for weak-profit sub-categories; introduce discount guardrails; prepare for peak months; and retrain the model regularly with richer business features.

5. Submission Contents

The project folder contains the dataset, analysis notebook, Excel dashboard/workbook, charts, Power BI-ready CSV exports and this report. A native .pbix file must be created/saved in Power BI Desktop.